

中国海洋大学全日制本科课程期末考试试卷

2026 年 Spring 季学期 考试科目: Statistical Modelling and Inference

学院: 海德学院

试卷类型: SAMPLE EXAM 2 SOLUTIONS 命题人: Thomas Shang 课程教师:
Dr. Waqar Hafeez

Examination instructions: This is a closed-book sample examination. A non-programmable calculator is permitted. Show all working. Formulae and statistical quantities required for each question are provided in the question.

Question	1	2	3	4	5	Total
Marks	20	20	20	20	20	100

Name: _____ Student ID: _____

Major/Year: _____ Seat Number: _____

Examination Room: _____ Instructor: _____

Sample Examination 2: Reference Solutions

Five questions, 20 marks each

End of cover page. Begin Question 1 on the next page.

Question 1 [20 marks total]

Combining two unbiased estimators, bias, variance, and MSE

Suppose $E(\hat{\theta}_1) = E(\hat{\theta}_2) = \theta$, $\text{Var}(\hat{\theta}_1) = 9$, and $\text{Var}(\hat{\theta}_2) = 16$. Define

$$\hat{\theta}_a = a\hat{\theta}_1 + (1 - a)\hat{\theta}_2.$$

For an estimator $\hat{\theta}$ of θ ,

$$\text{Bias}(\hat{\theta}) = E(\hat{\theta}) - \theta, \quad \text{MSE}(\hat{\theta}) = \text{Var}(\hat{\theta}) + \text{Bias}(\hat{\theta})^2.$$

For constants a and b ,

$$\text{Var}(aX + bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2ab \text{Cov}(X, Y).$$

If the estimators are independent, the covariance term is zero.

- (a) Prove that $\hat{\theta}_a$ is unbiased for any fixed a , and state its bias. [4 marks]
- (b) Assuming independence, find $\text{Var}(\hat{\theta}_a)$ and $\text{MSE}(\hat{\theta}_a)$. [5 marks]
- (c) Find the value of a that minimises the MSE under independence. [5 marks]
- (d) Now suppose $\text{Cov}(\hat{\theta}_1, \hat{\theta}_2) = 3$. Find the MSE-minimising value of a . [6 marks]

Solution

(a)

$$E(\hat{\theta}_a) = a\theta + (1 - a)\theta = \theta.$$

Therefore $\text{Bias}(\hat{\theta}_a) = 0$ for every fixed a .

(b) Under independence,

$$\text{Var}(\hat{\theta}_a) = 9a^2 + 16(1 - a)^2.$$

Since the bias is zero, $\text{MSE}(\hat{\theta}_a) = 9a^2 + 16(1 - a)^2$.

(c) Minimising the MSE is the same as minimising the variance. Differentiate:

$$\frac{d}{da} \{9a^2 + 16(1 - a)^2\} = 18a - 32(1 - a) = 50a - 32.$$

Hence $a = 32/50 = 0.64$. Equivalently,

$$a = \frac{\sigma_2^2}{\sigma_1^2 + \sigma_2^2} = \frac{16}{25} = 0.64.$$

(d) With covariance $c = 3$,

$$\text{MSE}(\hat{\theta}_a) = \text{Var}(\hat{\theta}_a) = 9a^2 + 16(1-a)^2 + 2a(1-a)3.$$

The optimal weight is

$$a = \frac{\sigma_2^2 - c}{\sigma_1^2 + \sigma_2^2 - 2c} = \frac{16 - 3}{9 + 16 - 6} = \frac{13}{19} \approx 0.684.$$

Question 2 [20 marks total]

Two-sample confidence interval and hypothesis test

A study compares mean weekly study hours for two independent groups of students.

	n	$\bar{y}$	s
Group A	36	18.4	4.8
Group B	49	15.9	5.6

Assume the samples are independent and large enough for a normal approximation.

For large independent samples,

$$SE = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}, \quad Z = \frac{(\bar{Y}_1 - \bar{Y}_2) - \Delta_0}{SE}.$$

Use $z_{0.025} = 1.96$, $z_{0.05} = 1.645$, and $\Phi(2.21) = 0.9864$.

- (a) Construct a 95% confidence interval for $\mu_A - \mu_B$. [8 marks]
- (b) Test $H_0 : \mu_A - \mu_B = 0$ against $H_a : \mu_A - \mu_B > 0$ at $\alpha = 0.05$. [7 marks]
- (c) Compute the p-value using the provided table value. [3 marks]
- (d) Interpret the result. [2 marks]

Solution

The estimated difference is $18.4 - 15.9 = 2.5$ and

$$SE = \sqrt{\frac{4.8^2}{36} + \frac{5.6^2}{49}} = \sqrt{0.64 + 0.64} = \sqrt{1.28} = 1.131.$$

(a)

$$2.5 \pm 1.96(1.131) = 2.5 \pm 2.217 = (0.283, 4.717).$$

(b)

$$Z_{\text{obs}} = \frac{2.5}{1.131} = 2.21.$$

Since $2.21 > 1.645$, reject H_0 .

(c)

$$p = P(Z \geq 2.21) = 1 - 0.9864 = 0.0136.$$

(d) The data provide evidence that Group A studies more hours per week on average than Group B.

Question 3 [20 marks total]

Multiple linear regression and least squares

Six students are observed. The response y is exam score, x_1 is weekly study hours, and x_2 is number of practice quizzes completed.

i	1	2	3	4	5	6
y_i	54	58	65	70	78	82
x_{1i}	1	2	3	4	5	6
x_{2i}	0	1	1	2	3	3

Fit $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon$.

$$(X^\top X)\hat{\beta} = X^\top y, \quad X^\top X = \begin{pmatrix} 6 & 21 & 10 \\ 21 & 91 & 44 \\ 10 & 44 & 24 \end{pmatrix}, \quad X^\top y = \begin{pmatrix} 407 \\ 1532 \\ 735 \end{pmatrix}.$$

- (a) Write the normal equations. [5 marks]
- (b) Solve for $\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2$. [9 marks]
- (c) Give the fitted regression equation. [3 marks]
- (d) Predict the score for $x_1 = 5$ and $x_2 = 2$. [3 marks]

Solution

The equations are

$$6b_0 + 21b_1 + 10b_2 = 407,$$

$$21b_0 + 91b_1 + 44b_2 = 1532,$$

$$10b_0 + 44b_1 + 24b_2 = 735.$$

Solving gives

$$\hat{\beta}_0 = 48.862, \quad \hat{\beta}_1 = 5.966, \quad \hat{\beta}_2 = -0.879.$$

Thus

$$\hat{y} = 48.862 + 5.966x_1 - 0.879x_2.$$

For $x_1 = 5$, $x_2 = 2$,

$$\hat{y} = 48.862 + 5.966(5) - 0.879(2) = 76.934.$$

Question 4 [20 marks total]

Fitted values and residual properties in simple linear regression

In the simple linear regression model

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i,$$

assume $E(Y_i) = \beta_0 + \beta_1 X_i$, $\text{Var}(Y_i) = \sigma^2$, and independent observations. Let

$$\hat{Y}_0 = \hat{\beta}_0 + \hat{\beta}_1 X_0, \quad \hat{e}_i = Y_i - \hat{Y}_i.$$

You may use

$$\begin{aligned} E(\hat{\beta}_0) &= \beta_0, \quad E(\hat{\beta}_1) = \beta_1, \quad \text{Var}(\hat{\beta}_0) = \sigma^2 \left(\frac{1}{n} + \frac{\bar{X}^2}{S_{xx}} \right), \\ \text{Var}(\hat{\beta}_1) &= \frac{\sigma^2}{S_{xx}}, \quad \text{Cov}(\hat{\beta}_0, \hat{\beta}_1) = -\frac{\sigma^2 \bar{X}}{S_{xx}}. \end{aligned}$$

- (a) Prove $E(\hat{Y}_0) = \beta_0 + \beta_1 X_0$. [5 marks]
- (b) Derive $\text{Var}(\hat{Y}_0) = \sigma^2 (1/n + (X_0 - \bar{X})^2/S_{xx})$. [9 marks]
- (c) Prove $\sum_{i=1}^n \hat{e}_i = 0$. [4 marks]
- (d) State why fitted-value variance is smallest at $X_0 = \bar{X}$. [2 marks]

Solution

(a)

$$E(\hat{Y}_0) = E(\hat{\beta}_0) + X_0 E(\hat{\beta}_1) = \beta_0 + \beta_1 X_0.$$

(b)

$$\begin{aligned} \text{Var}(\hat{Y}_0) &= \text{Var}(\hat{\beta}_0 + X_0 \hat{\beta}_1) \\ &= \text{Var}(\hat{\beta}_0) + X_0^2 \text{Var}(\hat{\beta}_1) + 2X_0 \text{Cov}(\hat{\beta}_0, \hat{\beta}_1) \\ &= \sigma^2 \left(\frac{1}{n} + \frac{\bar{X}^2}{S_{xx}} \right) + X_0^2 \frac{\sigma^2}{S_{xx}} - 2X_0 \frac{\sigma^2 \bar{X}}{S_{xx}} \\ &= \sigma^2 \left(\frac{1}{n} + \frac{(X_0 - \bar{X})^2}{S_{xx}} \right). \end{aligned}$$

(c) Since $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$,

$$\sum \hat{e}_i = \sum (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) = n\bar{Y} - n\hat{\beta}_0 - n\hat{\beta}_1 \bar{X} = 0.$$

(d) The variance expression contains $(X_0 - \bar{X})^2$, which is minimised at $X_0 = \bar{X}$.

Question 5 [20 marks total]

Normal MLE and Fisher information

Let $Y_1, \dots, Y_n \stackrel{iid}{\sim} N(\mu, \sigma^2)$.

The normal density is

$$f(y; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ -\frac{(y - \mu)^2}{2\sigma^2} \right\}.$$

For parameter vector (μ, v) with $v = \sigma^2$,

$$I(\mu, v) = -E\{H(\mu, v)\},$$

where H is the Hessian matrix of the log-likelihood.

- (a) Assume $v = \sigma^2$ is known. Derive the MLE of μ . [5 marks]
- (b) Now assume both μ and $v = \sigma^2$ are unknown. Write $\ell(\mu, v)$. [4 marks]
- (c) Derive the score components $\partial\ell/\partial\mu$ and $\partial\ell/\partial v$. [5 marks]
- (d) State the Fisher information matrix for (μ, v) . [4 marks]
- (e) Explain why the cross-information term is zero. [2 marks]

Solution

- (a) With v known,

$$\ell(\mu) = -\frac{n}{2} \log(2\pi v) - \frac{1}{2v} \sum (y_i - \mu)^2.$$

The score is

$$\frac{\partial\ell}{\partial\mu} = \frac{1}{v} \sum (y_i - \mu).$$

Setting it to zero gives $\hat{\mu} = \bar{y}$.

- (b)

$$\ell(\mu, v) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log v - \frac{1}{2v} \sum_{i=1}^n (y_i - \mu)^2.$$

- (c)

$$\frac{\partial\ell}{\partial\mu} = \frac{1}{v} \sum (y_i - \mu), \quad \frac{\partial\ell}{\partial v} = -\frac{n}{2v} + \frac{1}{2v^2} \sum (y_i - \mu)^2.$$

(d) The Fisher information matrix is

$$I(\mu, v) = \begin{pmatrix} n/v & 0 \\ 0 & n/(2v^2) \end{pmatrix}.$$

(e) The cross second derivative is proportional to $\sum(Y_i - \mu)$, whose expectation is zero.

End of examination questions.

Check that you have attempted all five questions.